Assignment No. 3

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Section	"A"
Deppt	Computer Science
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Subject	Discrete Structure

Question No (1):-

Part (a):-

Total members = 14

In how many ways can a group of seven be chosen to work on a project.

=> To the calculate the number of ways we will use the combination formula.

$$c(n,k) = \frac{n!}{(n-k)!\,k!}$$

$n = 14$; $k = 7$

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②

$$c(14,7) = \frac{14!}{(14-7)!\,7!}$$

$$c(14,7) = \frac{14!}{7!\,7!}$$

$$c(n,k) = \frac{14\times13\times12\times11\times13\times12\times11\times10\times9\times8\times\cancel{7!}}{7\times6\times5\times4\times3\times2\times1\times\cancel{7!}}$$

$$c(n,k) = \frac{19208}{5040}$$

$$c(n,k) = 3432$$

Part (b):- Suppose eight team members are women and six are men

i) How many groups of seven can be chosen that contain four women and three men?

Total members = 14

Men = 6 = n

women = 8 = n

To be chosen;

Men = 3 = k

Women = 4 = k

$$c(8,4)\times c(6,3) = \frac{8!}{(8-4)\,4!}\times\frac{6!}{3!\,3!}$$

$$c(8,4)\times c(6,3) = \frac{8\times7\times6\times5\times\cancel{4!}}{\cancel{4!}\times4\times3\times2\times1}\times\frac{\cancel{6}^{2}\times5\times\cancel{4}\times3}{\cancel{3}\times2\times\cancel{1}\times\cancel{3!}}$$

③ $c(8,4) \times c(6,3) = 70 \times 20$

So there are 1400 ways

Ans

(ii) Number of ways to choose a group of seven with at least one man.

Total ways to choose a group of seven
ways to choose a group with no men.

$c(14,7) - c(8,7) = 3432 - 8$
$= 3424$

So there are 3424 ways to choose group.

(iii) Number of ways to choose a group of seven with at least three women:

We consider the different possibilities for the number of women in the group.

0, 1, 2, 3 women we calculate for each case and then sum it.

④ **for 0.**

There are no ways to chose 0 from 8 members.

for 1 women:-

$$c(8,1) \times c(6,6) = \frac{8!}{(8-1)!} \times \frac{6!}{(6-6)!\,6!}$$

$$c(8,1) \times c(6,6) = \frac{8 \times \cancel{7!}}{\cancel{7!}} \times 1$$

$$c(8,1) \times c(6,6) = 8 \text{ ways}$$

for 2 women.

$$c(8,2) \times c(6,5) = \frac{8!}{(8-2)!\,2!} \times \frac{6!}{(6-5)!\,5!}$$

$$= \frac{\overset{4}{\cancel{8}} \times 7 \times \cancel{6!}}{\cancel{6!} \times 2 \times 1} \times \frac{6 \times \cancel{5!}}{1!\,\cancel{5!}}$$

$$c(8,2) \times c(6,5) = 168 \text{ ways}$$

for 3 women

$$c(8,3) \times c(6,4) = \frac{8!}{(8-3)!\,3!} \times \frac{6!}{(6-4)!\,4!}$$

$$= \frac{\overset{4}{\cancel{8}} \times 7 \times \overset{2}{\cancel{6}} \times \cancel{5!}}{\cancel{5!} \cdot \cancel{3!} \times \cancel{2} \times 1} \times \frac{\overset{3}{\cancel{6}} \times 5 \times \cancel{4!}}{2 \times \cancel{4!} \times \cancel{3 \times 2} \times 1}$$

$$= 56 \times 15$$

$$c(8,3) \times c(6,4) = 840 \text{ ways}$$

So Total combination for atmost 3 women are

$$= 0 + 8 + 168 + 84$$

$$= \boxed{1016 \text{ Total Combination.}}$$

5) Part (c):

No of ways to choose a group of seven when two members refuse to work together.

= Total ways to choose a group of 7
= ways to choose a group with both member included.

$c(14,7) = 3432$ already solved.

for

$c(13,6) = \frac{13!}{(13-5)!6!} = \frac{12 \times 13 \times 10 \times 9 \times 10 \times 9 \times 8 \times 7!}{7! \times 5 \times 4 \times 3 \times 2}$

$c(13,6) = \underline{1716}$ ways when two refuse to work.

Total $= 3432 - 1716 = 1716$ ways

Part. (D):

No of ways to choose a group of seven when two members insist on working together or not at all.

= No of ways of both included + No of ways of both excluded.

$c(14,7) = 3432$ already solved.

7) 6) for next part both members are included.

$n = 12, \quad k = 5$

$$C(12,5) = \frac{12!}{(12-5)!\,5!} = \frac{12 \times 11 \times 10 \times 9 \times 8 \times \cancel{7!}}{\cancel{7!} \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$C(12,7) = 792$$

Total ways $= 792 + 792 = \boxed{1584}$ Ans

7) Question (2):-

How many ways can 12 students be divided into 3 groups with 4 student in each group so that.

(part (i))

one group studies English; one history; and one Maths.

Solution

1) For English

$n = 12, \quad k = 4$

$$c(12,4) = \frac{12!}{(12-4)!\,4!} = \frac{12 \times 11 \times 10 \times 9 \times \cancel{8!}}{\cancel{8!} \times 4 \times 3 \times 2 \times 1}$$

$$c(12,4) = 495$$

2) For History:

$n = 8; \quad k = 4$

$$c(8,4) = \frac{8!}{4!\,(8-4)!} = \frac{8 \times 7 \times 6 \times 5 \times \cancel{4!}}{4 \times 3 \times 2 \times 1 \times \cancel{4!}}$$

$$c(8,4) = 70$$

3) The remaining four automatically form the mathematics group.

$$\boxed{\text{Total ways} = 495 \times 70 = 34650}$$ Ans

8) Part (ii)

All the groups study mathematics.

We need to chose 4 students for each group without considering the objects.

Total ways = $C(12,4) \times C(8,4) \times C(4,4)$.

$= 34650$ ways

which is again the same as previous one.

Question No (3.):-

To find the number of ways to partition a class of ten students into four terms.

Two teams of two students and two teams of 3 students.

* Solution.

Choose two students for A1.

$A = 10, \quad k = 2$

$C(10,2) = \frac{10!}{(10-2)!\,2!} = \frac{\cancel{10}^{5} \times 9 \times \cancel{8!}}{\cancel{8!} \times \cancel{2} \times 1}$

$C(10,2) = 5 \times 9 = \boxed{45 \text{ ways}}$

9) choose 2 students for A 2:-

$n = 8, k = 2$

$c(8,2) = \frac{8!}{2!(8-2)!} = \frac{\cancel{8}^{4} \times 7 \times \cancel{6!}}{2 \times 1 \times \cancel{6!}}$

$c(8,2) = 28$ ways

choose 3 students for B 1:-

$n = 6, k = 3$

$c(6,3) = \frac{6!}{(6-3)!3!} = \frac{\cancel{6}^{2} \times 5 \times \cancel{4} \times \cancel{3!}}{\cancel{3} \times \cancel{2!}\cancel{3!}}$

$c(6,3) = 20$

choose last students for B 2:

$n = 3, k = 3$

$c(3,3) =$ The remaining form B 2 team.

$$\boxed{\text{Total ways} = 45 \times 28 \times 20 = 25200 \text{ ways}}$$

Question (4) :- To find number of possible arrangments for C1, C2, C3 C4, C5 and C6.

Total number of people = 20 people

1) Choose 3 people for C1:

$c(20,3) = \frac{20!}{(20-3)!\,3!} = \frac{20 \times 19 \times 18 \times \cancel{17!}}{\cancel{17!} \times 3 \times 2 \times 1}$

10) $c(20,3) = 1140$.

2) Choose 3 people for C2:

$$c(17,3) = \frac{17!}{3!(17-3)!} = 680.$$

3) Choose 4 people for C3:

$$c(14,4) = \frac{14!}{4!(14-4)!} = 1001$$

4) Choose 4 people for C4:

$$c(10,4) = \frac{10!}{4!(10-4)!} = 210$$

5) Choose 2 people to C5:

$$c(6,2) = \frac{6!}{2!(6-2)!} = 15$$

6) The remaining four people form C6.

Hence the total number of ways possible are.

$$\boxed{\begin{aligned} \text{Total ways} &= 1140 \times 680 \times 1001 \times 210 \times 15 \\ &= 2{,}444{,}321{,}880{,}000 \end{aligned}}$$

Ans.